

Lambda calculus

Exercise sheet 5 (solutions)

Exercise 5.1. Let $ID = \lambda x.x$ and $\Omega = (\lambda x.xx)(\lambda x.xx)$. Reduce the following terms to normal form:

- (a) $(\lambda xy.yx) z ID$
- (b) $(\lambda x.\lambda y.xy) y$
- (c) $TRUE ID \Omega$
- (d) $\bar{1} \bar{2} \bar{3}$

Solution. (a) The first application supplies z ; the second applies the identity function to it:

$$(\lambda x.\lambda y.yx) z ID \longrightarrow_{\beta} (\lambda y.yz) ID \longrightarrow_{\beta} ID z \longrightarrow_{\beta} z.$$

- (b) Rename the inner parameter to z so that the argument y remains free:

$$(\lambda x.\lambda y.xy) y \equiv_{\alpha} (\lambda x.\lambda z.xz) y \longrightarrow_{\beta} \lambda z.yz.$$

- (c) $TRUE$ selects ID , discarding Ω without evaluating it:

$$TRUE ID \Omega \longrightarrow_{\beta}^* ID = \lambda x.x.$$

- (d) $\bar{1}$ applies $\bar{2}$ once to $\bar{3}$. The term $\bar{3}f$ applies f three times; $\bar{3}(\bar{3}f)$ applies this new function three times, giving $3 \cdot 3 = 9$ applications of f . Thus $\bar{2} \bar{3}$ triples the repetition count twice. We show this formally. After

$$\bar{1} \bar{2} \bar{3} \longrightarrow_{\beta}^* \bar{2} \bar{3},$$

each arrow below is one beta reduction:

$$\begin{aligned} \bar{2} \bar{3} &\longrightarrow_{\beta} \lambda f.\bar{3}(\bar{3}f) \\ &\longrightarrow_{\beta} \lambda f.\lambda x.(\bar{3}f)((\bar{3}f)((\bar{3}f)x)) \\ &\longrightarrow_{\beta} \lambda f.\lambda x.(\bar{3}f)((\bar{3}f)((\lambda u.f^3u)x)) \\ &\longrightarrow_{\beta} \lambda f.\lambda x.(\bar{3}f)((\bar{3}f)(f^3x)) \\ &\longrightarrow_{\beta} \lambda f.\lambda x.(\bar{3}f)((\lambda u.f^3u)(f^3x)) \\ &\longrightarrow_{\beta} \lambda f.\lambda x.(\bar{3}f)(f^6x) \\ &\longrightarrow_{\beta} \lambda f.\lambda x.(\lambda u.f^3u)(f^6x) \\ &\longrightarrow_{\beta} \lambda f.\lambda x.f^9x = \bar{9}. \end{aligned}$$

□

Exercise 5.2. Define as lambda terms:

- (a) disjunction;
- (b) implication;
- (c) exclusive disjunction.

Solution. Disjunction returns true if p is true, and otherwise returns q . Implication returns q if p is true, and otherwise returns true. Exclusive disjunction negates q if p is true, and otherwise returns q :

$$\begin{aligned}\text{OR} &= \lambda p. \lambda q. p p q, \\ \text{IMPLIES} &= \lambda p. \lambda q. p q \text{ TRUE}, \\ \text{XOR} &= \lambda p. \lambda q. p (\text{NOT } q) q.\end{aligned}$$

For Boolean inputs, disjunction can also be expressed using De Morgan's law:

$$\text{OR } A B =_{\beta} \text{NOT}(\text{AND}(\text{NOT } A)(\text{NOT } B)).$$

□

Exercise 5.3. Define the following functions for Church numerals:

- (a) zero test: $\bar{0} \mapsto \text{TRUE}$, $\bar{0} \neq \bar{n} \mapsto \text{FALSE}$;
- (b) parity test: $\overline{2k} \mapsto \text{TRUE}$, $\overline{2k+1} \mapsto \text{FALSE}$;
- (c) exponentiation: $\bar{n}, \bar{m} \mapsto \overline{n^m}$.

Solution. (a) For ISZERO, iterate a function that always returns FALSE, starting from TRUE. Zero iterations leave TRUE unchanged; any positive number returns FALSE:

$$\text{ISZERO} = \lambda n. n (\lambda x. \text{FALSE}) \text{ TRUE}.$$

- (b) For EVEN, negate the current Boolean at each iteration, starting from TRUE. An even number of negations returns TRUE; an odd number returns FALSE:

$$\text{EVEN} = \lambda n. n \text{ NOT TRUE}.$$

- (c) For EXP, multiply by n exactly m times, starting from 1. Zero iterations return $\bar{1}$, including for 0^0 :

$$\text{EXP} = \lambda n. \lambda m. m (\text{MULT } n) \bar{1}.$$

□

Exercise 5.4. Give explicit definitions, i.e. without relying on previously defined terms, for:

- (a) addition;
- (b) multiplication;
- (c) * exponentiation (assuming η -equivalence, i.e. the rule $\lambda x.Mx \equiv_\eta M$, provided x does not occur free in M).

Solution. (a) For ADD, apply f first n times and then m times:

$$\text{ADD} = \lambda m.\lambda n.\lambda f.\lambda x.mf(nfx).$$

- (b) For MULT, repeat the function nf , which applies f n times, a total of m times:

$$\text{MULT} = \lambda m.\lambda n.\lambda f.m(nf).$$

- (c) For EXP, each application of $\bar{n}$ multiplies the repetition count by n , so $\bar{m}\bar{n}$ gives $\bar{n^m}$ when $m > 0$:

$$\text{EXP} = \lambda n.\lambda m.mn.$$

To see this in general, unfold the first numeral:

$$\bar{m}\bar{n} = (\lambda g.\lambda f.g^m f)\bar{n} \longrightarrow_\beta \lambda f.\underbrace{\bar{n}(\bar{n}(\dots(\bar{n}f)\dots))}_{m \text{ times}}.$$

Note that $\bar{n}(\bar{n}(\dots(\bar{n}f)\dots))$ reduces to $\lambda x.[\dots]$, and with $\lambda f.$ in front it becomes a numeral—for $m > 0$ we have:

$$\bar{m}\bar{n} \longrightarrow_\beta^* \lambda f.\lambda x.f^{n^m}x = \bar{n^m}.$$

The case $m = 0$ is slightly less elegant:

$$\bar{0}\bar{n} = (\lambda g.\lambda f.f)\bar{n} \longrightarrow_\beta \lambda f.f.$$

The argument $\bar{n}$ is discarded because g does not occur in the body $\lambda f.f$. The result is already in beta normal form, but is not the Church numeral $\bar{1}$. For that, choose x distinct from f and apply the eta rule backwards to the body f :

$$f \equiv_\eta \lambda x.fx, \quad \text{hence} \quad \lambda f.f \equiv_\eta \lambda f.(\lambda x.fx) = \lambda f.\lambda x.fx = \bar{1}.$$

Here we take $0^0 = 1$, as in Exercise 5.3. To obtain the exact Church numeral using beta reduction alone for all m, n , supply both parameters explicitly:

$$\text{EXP} = \lambda n.\lambda m.\lambda f.\lambda x.mnf x.$$

□